



Creating immune cells from iPSCs to investigate neurodegenerative disease

Freddy Gibson, BSc, PhD
Technical Specialist
Cellular Services, Cellular Operations



Who am I?



What is neurodegenerative disease?



What is neurodegenerative disease?

- Progressive loss of neurons
 - > movement issues
 - > cognitive decline
 - > death
- Irreversible + Incurable
- Symptoms vary depending on area region affected
 - > many different diseases, ie Alzheimer's
- Strongest risk factor -> age (65+)




Causes of neurodegenerative disease?

- Genetics
 - Many linked to specific gene mutations
 - CAG repeats -> Huntington's
- Protein misfolding
 - Lewy bodies -> Parkinson's
 - Amyloid beta + Tau -> Alzheimer's
 - Prions -> Kuru, BSE (mad cow disease)
- How mutations and protein misfolding cause disease is not well understood



Management, not cure

- Once neurons are gone they cannot be replaced
- Halt/slow decline
 - Clear protein aggregates
 - Protect remaining neurons
- Prevention
 - Immunotherapy, ie vaccine
 - Disrupt protein aggregate formation
 - Lifestyle



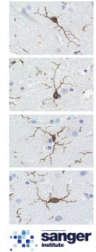
Why don't we have a better understanding?

- Mechanism of disease not well understood
- Research difficult to do
 - Slow progressive disease -> hard to study in humans
 - Low sample availability - brain tissue is hard to get!
 - Many types of brain cell that may influence disease
- Current research methods
 - Genetic studies, ie sequencing patients
 - Cell models, ie donated tissue, neuron-like cell lines
 - *In vivo* models, ie mice



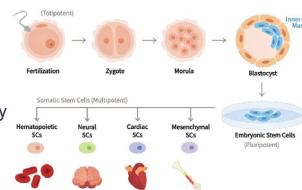
Our focus -> use a different research model

- Aim: study mutation of specific genes in microglia, an immune brain cell linked to ND risk
- Problem: limited sample availability
 - Isolate from patients: low amount, highly variable
 - Cell line: act differently to real microglia
- Our solution: make them in the lab using human induced pluripotent stem cells (iPSCs)



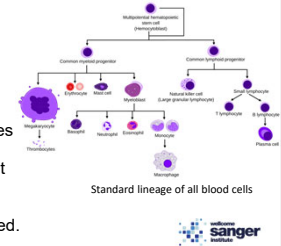
What are stem cells?

- Divide -> more stem cells + a new cell type
- Make (differentiate into) all the different types of cell in the body
- Potency: how many different types of cell types it can make
- Induced: by feeding any cell the right signals we can convert it back into a stem cell

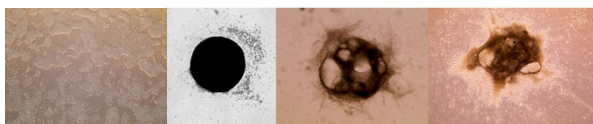


Generating microglia at scale

- Two step process
 - iPSCs -> myeloid precursors
 - Myeloid precursors -> microglia
- Why myeloid precursors?
 - Can make them in large quantities (up to 1.5 billion!)
 - Can be maintained in the lab, but cannot freeze
 - By adding new factors can be turned into microglia when needed.



How do we do it?

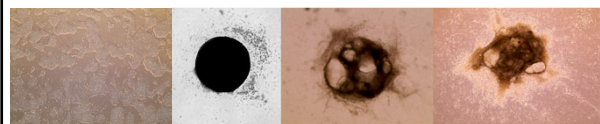


iPSC culture ~1 week Embryoid body development ~1 week Transfer to and maturation in Factory Flasks ~7 weeks

- Whole process up to 3 months long!



How do we do it?

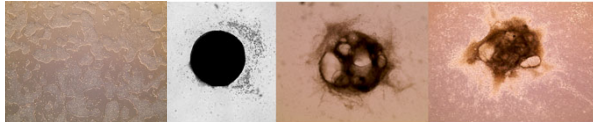


iPSC culture ~1 week Embryoid body development ~1 week Transfer to and maturation in Factory Flasks ~7 weeks

- iPSC = human cell line derived here at the Sanger
- can be stored long-term in LN₂
- grow on flat 2D plate in colonies



How do we do it?



iPSC culture
~1 week

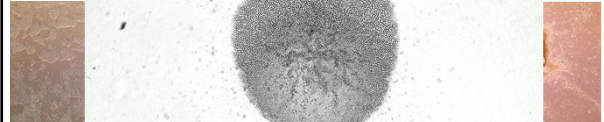
**Embroid body
development**
~1 week

Transfer to and maturation in Factory Flasks
~7 weeks

- transfer iPSCs to curved U-bottom plate
- add new factors the 'media'
- grow into 3D structures over a week



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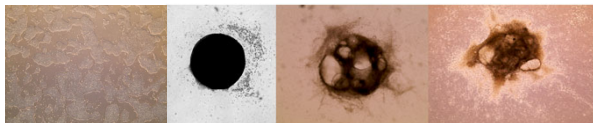
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~1

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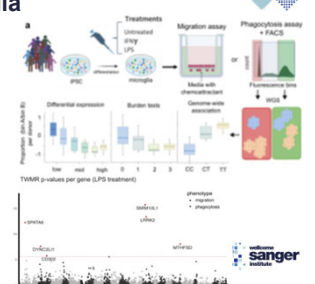
Transfer to and maturation in Factory Flasks
~7 weeks

- transfer EBs to larger flask and embed in gelatin
- develop a 'skirt' of cells spreading out from the EB
- after 2 weeks can begin harvesting mature MPs



Investigating microglia

- Expose to stimuli/mutate genes of interest
- Test how cells look and act
- DNA and RNA sequencing



What do I do?

- Technical Specialist within the Cellular Operations department
- Cellular Operations
 - Support academic teams with large scale projects/pipelines
 - In-house 'out-sourcing'
- Multiple projects ongoing -> Myelogenesis + Skin Organoids
- Other responsibilities
 - Health and safety
 - Ergonomics



Key skills

- Problem solving
 - Get given problem -> find a solution
 - Process goes wrong -> find out why + how to avoid in future
- Perseverance
 - When things go wrong stay positive!
- Team work and communication
 - Can do much more together than alone
 - Use each others strengths and knowledge
- Attention to detail
 - Small changes can have large effects
 - Record keeping is key



**Thanks for listening
Any questions?**

